

electric networks by using a CAS (computer algebra system) approach. SimCAS uses symbolic equations to define components and solves systems without using matrix. It uses the Taylor Expansion Diagrams theory.

SimCAS can handle non-linear DC, non-linear transient and linear AC analysis. It is built with wxWidget and uses the framework library wxAUI. It is delivered in two parts: GUI source code based on wxWidget set of classes and mathematical kernel source code that is tool-set independent. It uses standard and built-in C++ classes.

CASPOC

CASPOC is especially designed for simulation of power electronics and electrical drives. Using CASPOC you can model power electronics and electrical machines, loads and controls in one multilevel model. This multilevel model includes a circuit level for modeling of switched-mode power supplies, a component level for modeling of electrical machines/loads and a system level for modeling of control algorithms.

The advantages of CASPOC include fast simulation of switched-mode power supplies without convergence problems, fast models for diode, SCR, GTO and semiconductor switches, special block-diagram components for power electronics and drive simulations.

It also features modeling of control algorithms using 'C' or Pascal, schematic editor, system and component models, and waveform viewer.

Electric

Electric VLSI Design System is an EDA tool written by Steven M. Rubin. It is used to draw schematics and do integrated circuit (IC) layout. It can also handle hardware description languages such as VHDL and Verilog.

It has a powerful way of doing IC layout. The system considers integrated circuits to be composed of nodes and arcs. Nodes are circuit elements such as transistors and contacts. Arcs connect the nodes. This style of design is quite different from typical IC layout systems, such as Magic and Cadence, which do IC layout by manipulating polygons on different layers of the wafer.

This powerful design style makes it possible for the designer to run Layout Versus Schematic (LVS) on layouts that are not yet design rule check (DRC) clean. This is extremely useful during the design process. It also makes it possible for the design tool to back-annotate wire-lengths from a layout to a schematic even if the layout is not yet DRC clean. Finally, because layouts are stored internally as a graph (rather than 'paint'), the LVS check is exceptionally fast—often running in under a second on full-chip designs (such as the recent Marina chip). The ability to run LVS near-instantaneously on large designs has a big impact on ease of design.

One added advantage of the nodes-and-arcs view of a circuit is that it is possible to add layout constraints to the arcs. There are only a small set of these constraints, but with careful placement, the circuit

can be programmed to stay properly connected when physical changes are made.

Electric incorporates many synthesis and analysis tools such as built-in design rule checkers and two interfaces to industry tools, electrical rule checking, a well/substrate checker and an antenna rule checker, two built-in simulators and interfaces to over a dozen industry tools (Spice, Verilog, etc), and five different routers for various special purposes.

It also has a PLA generator, a cell generator, a pad frame generator and a ROM generator. It consists of a tool for analysing the logical effort of a circuit and resising gates, LVS (layout vs schematic) tool to compare any two representations of a circuit, and I/O capability to read and write circuit descriptions in many formats, including CIF, GDS, EDIF, DXF and VHDL. It can also interface with other systems such as Eagle, Pads, ECAD and Sue.

naga EDA

naga EDA provides useful electronic design tools in C++ and, especially, Python. The current release contains naga. Verilog—a Verilog parser. The idea of naga EDA is to provide Open Source EDA tools to grant users complete control over the design process so as to ease the daily re-developing tasks. The core classes and algorithms are written in C++ to maximise performance. There is a rich Python access interface to C++ parts providing user-friendly scripting environment. ●

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